

Mitogenome Codon Review

Generated: 2026-05-27 | **Records:** 30 | **CDS analyzed:** 388

Command: `/home/dmacguig/Documents/GitHub/GenBankMitoReview/summarize_codons.py examples/asteroidea.gb -o examples/asteroidea --noncoding-stats`

Input: `examples/asteroidea.gb`

Full per-CDS data table: `examples/asteroidea.codons.tsv`

Genetic Code Summary

Table 9: Echinoderm Mitochondrial — 30 record(s) - Canonical start codons: `ATG` · `GTG` - Canonical stop codons: `TAA` · `TAG`

Codon Usage by Gene

388 CDS analyzed across 13 gene group(s). Canonical status uses each record's `transl_table` genetic code.

atp6

30 CDS · 30 complete · 0 partial

Length (bp): min=687 · max=693 · mean=692.3 · SD=1.8

Start codons

Codon	n	%	Status
ATG	30	100.0%	ok

Non-canonical start: 0/30 (0.0%)

Stop codons

Codon	n	%	Status
TAA	28	93.3%	ok
T (polyA)	1	3.3%	polyA
TAG	1	3.3%	ok

polyA-completed (T/TA → TAA): 1/30 (3.3%)

Non-canonical stop: 0/30 (0.0%)

atp8

29 CDS · 29 complete · 0 partial

Length (bp): min=162 · max=177 · mean=167.2 · SD=3.2

Start codons

Codon	n	%	Status
ATG	29	100.0%	ok

Non-canonical start: 0/29 (0.0%)

Stop codons

Codon	n	%	Status
TAA	28	96.6%	ok
TAG	1	3.4%	ok

Non-canonical stop: 0/29 (0.0%)

cob

30 CDS · 30 complete · 0 partial

Length (bp): min=1138 · max=1158 · mean=1138.7 · SD=3.6

Start codons

Codon	n	%	Status
ATG	30	100.0%	ok

Non-canonical start: 0/30 (0.0%)

Stop codons

Codon	n	%	Status
T (polyA)	29	96.7%	polyA
TAA	1	3.3%	ok

polyA-completed (T/TA → TAA): 29/30 (96.7%)

Non-canonical stop: 0/30 (0.0%)

cox1

30 CDS · 30 complete · 0 partial

Length (bp): min=1551 · max=1557 · mean=1553.8 · SD=0.9

Start codons

Codon	n	%	Status
ATG	30	100.0%	ok

Non-canonical start: 0/30 (0.0%)

Stop codons

Codon	n	%	Status
TAA	21	70.0%	ok
TA (polyA)	5	16.7%	polyA
TAG	3	10.0%	ok
T (polyA)	1	3.3%	polyA

polyA-completed (T/TA → TAA): 6/30 (20.0%)

Non-canonical stop: 0/30 (0.0%)

cox2

30 CDS · 30 complete · 0 partial

Length (bp): min=688 · max=702 · mean=689.2 · SD=2.7

Start codons

Codon	n	%	Status
ATG	30	100.0%	ok

Non-canonical start: 0/30 (0.0%)

Stop codons

Codon	n	%	Status
T (<i>polyA</i>)	19	63.3%	polyA
TAA	9	30.0%	ok
TAG	2	6.7%	ok

polyA-completed (T/TA → TAA): 19/30 (63.3%)

Non-canonical stop: 0/30 (0.0%)

cox3

30 CDS · 30 complete · 0 partial

Length (bp): min=780 · max=783 · mean=782.8 · SD=0.7

Start codons

Codon	n	%	Status
ATG	30	100.0%	ok

Non-canonical start: 0/30 (0.0%)

Stop codons

Codon	n	%	Status
TAA	30	100.0%	ok

Non-canonical stop: 0/30 (0.0%)

nad1

30 CDS · 28 complete · 2 partial

Length (bp): min=948 · max=981 · mean=977.6 · SD=5.9

Start codons

Codon	n	%	Status
GTG	17	60.7%	ok
ATG	11	39.3%	ok

Non-canonical start: 0/28 (0.0%)

Stop codons

Codon	n	%	Status
TAG	15	53.6%	ok
TAA	11	39.3%	ok
T (polyA)	2	7.1%	polyA

polyA-completed (T/TA → TAA): 2/28 (7.1%)

Non-canonical stop: 0/28 (0.0%)

2 partial CDS (truncated/fuzzy ends — not assessed)

Record	Species	Start codon	Start flag	Stop codon	Stop flag
NC_054231.1	Euretaster insignis	TTG	partial	TAG	ok
NC_085630.1		TTG	partial	TAG	ok

Record	Species	Start codon	Start flag	Stop codon	Stop flag
	Xyloplax princealberti				

nad2

30 CDS · 30 complete · 0 partial

Length (bp): min=1062 · max=1095 · mean=1065.9 · SD=6.5

Start codons

Codon	n	%	Status
ATG	15	50.0%	ok
GTG	15	50.0%	ok

Non-canonical start: 0/30 (0.0%)

Stop codons

Codon	n	%	Status
TAA	23	76.7%	ok
TAG	7	23.3%	ok

Non-canonical stop: 0/30 (0.0%)

nad3

30 CDS · 12 complete · 18 partial

Length (bp): min=333 · max=351 · mean=350.3 · SD=3.3

Start codons

Codon	n	%	Status
ATG	8	66.7%	ok

Codon	n	%	Status
ATT	4	33.3%	NON-CANONICAL

Non-canonical start: 4/12 (33.3%)

Stop codons

Codon	n	%	Status
TAA	10	83.3%	ok
TAG	2	16.7%	ok

Non-canonical stop: 0/12 (0.0%)

18 partial CDS (truncated/fuzzy ends — not assessed)

Record	Species	Start codon	Start flag	Stop codon	Stop flag
NC_001627.1	Patiria pectinifera	ATT	partial	TAG	ok
NC_037943.1	Echinaster brasiliensis	ATT	partial	TAG	ok
NC_053361.1	Crossaster papposus	ATT	partial	TAG	ok
NC_054225.1	Pentaceraster mammillatus	ATT	partial	TAA	ok
NC_054226.1	Protoreaster nodosus	ATT	partial	TAA	ok
NC_054227.1	Ophidiaster granifer	ATT	partial	TAA	ok
NC_054228.1	Iconaster longimanus	ATT	partial	TAA	ok
NC_054229.1	Culcita novaeguineae	ATT	partial	TAA	ok
NC_054230.1	Anthenea aspera	ATT	partial	TAA	ok
NC_054231.1	Euretaster insignis	TTG	partial	TAA	ok
NC_063668.1	Asthenactis papyraceus	ATT	partial	TAA	ok
NC_063669.1	Zoroaster ophiactis	ATT	partial	TAA	ok

Record	Species	Start codon	Start flag	Stop codon	Stop flag
NC_081967.1	Leptychaster arcticus	ATT	partial	TAA	ok
NC_081981.1	Crossaster japonicus	ATT	partial	TAG	ok
NC_083191.1	Poraniopsis inflata	ATT	partial	TAA	ok
NC_085630.1	Xyloplax princealberti	ATT	partial	TAA	ok
NC_087784.1	Henricia reniossa	ATT	partial	TAA	ok
NC_087785.1	Henricia sanguinolenta	ATT	partial	TAA	ok

nad4

30 CDS · 30 complete · 0 partial

Length (bp): min=1377 · max=1386 · mean=1382.5 · SD=1.6

Start codons

Codon	n	%	Status
ATG	30	100.0%	ok

Non-canonical start: 0/30 (0.0%)

Stop codons

Codon	n	%	Status
TAA	23	76.7%	ok
TAG	7	23.3%	ok

Non-canonical stop: 0/30 (0.0%)

nad41

30 CDS · 8 complete · 22 partial

Length (bp): min=288 · max=321 · mean=297.3 · SD=4.7

Start codons

Codon	n	%	Status
ATT	4	50.0%	NON-CANONICAL
ATG	3	37.5%	ok
ATC	1	12.5%	NON-CANONICAL

Non-canonical start: 5/8 (62.5%)

Stop codons

Codon	n	%	Status
TAA	8	100.0%	ok

Non-canonical stop: 0/8 (0.0%)

22 partial CDS (truncated/fuzzy ends — not assessed)

Record	Species	Start codon	Start flag	Stop codon	Stop flag
NC_001627.1	Patiria pectinifera	ATT	partial	TAA	ok
NC_037943.1	Echinaster brasiliensis	ATT	partial	TAA	ok
NC_041450.1	Styracaster yapensis	ATT	partial	TAA	ok
NC_042741.1	Pisaster ochraceus	ATT	partial	TAA	ok
NC_053361.1	Crossaster papposus	GTC	partial	TAA	ok
NC_054225.1	Pentaceraster mammillatus	ATT	partial	TAA	ok
NC_054226.1	Protoreaster nodosus	ATT	partial	TAA	ok
NC_054227.1		ATC	partial	TAA	ok

Record	Species	Start codon	Start flag	Stop codon	Stop flag
	Ophidiaster granifer				
NC_054228.1	Iconaster longimanus	ATC	partial	TAA	ok
NC_054229.1	Culcita novaeguineae	ATC	partial	TAA	ok
NC_054230.1	Anthenea aspera	ATC	partial	TAA	ok
NC_054231.1	Euretaster insignis	ATG	partial	TAA	ok
NC_063669.1	Zoroaster ophiactis	ATT	partial	TAA	ok
NC_063787.1	Coscinasterias acutispina	ATT	partial	TAA	ok
NC_071876.1	Ctenodiscus crispatus	ATT	partial	TAA	ok
NC_081967.1	Leptychaster arcticus	ATT	partial	TAA	ok
NC_081981.1	Crossaster japonicus	GTT	partial	TAA	ok
NC_083190.1	Sclerasterias satsumana	ATT	partial	TAA	ok
NC_083191.1	Poraniopsis inflata	ATT	partial	TAA	ok
NC_085630.1	Xyloplax princealberti	TTG	partial	TAA	ok
NC_087784.1	Henricia reniossa	ATC	partial	TAA	ok
NC_087785.1	Henricia sanguinolenta	ATC	partial	TAA	ok

nad5

29 CDS · 29 complete · 0 partial

Length (bp): min=1842 · max=1932 · mean=1903.0 · SD=18.3

Start codons

Codon	n	%	Status
ATG	28	96.6%	ok
GTG	1	3.4%	ok

Non-canonical start: 0/29 (0.0%)

Stop codons

Codon	n	%	Status
TAA	26	89.7%	ok
TAG	3	10.3%	ok

Non-canonical stop: 0/29 (0.0%)

nad6

30 CDS · 30 complete · 0 partial

Length (bp): min=474 · max=492 · mean=488.9 · SD=3.0

Start codons

Codon	n	%	Status
ATG	27	90.0%	ok
GTG	3	10.0%	ok

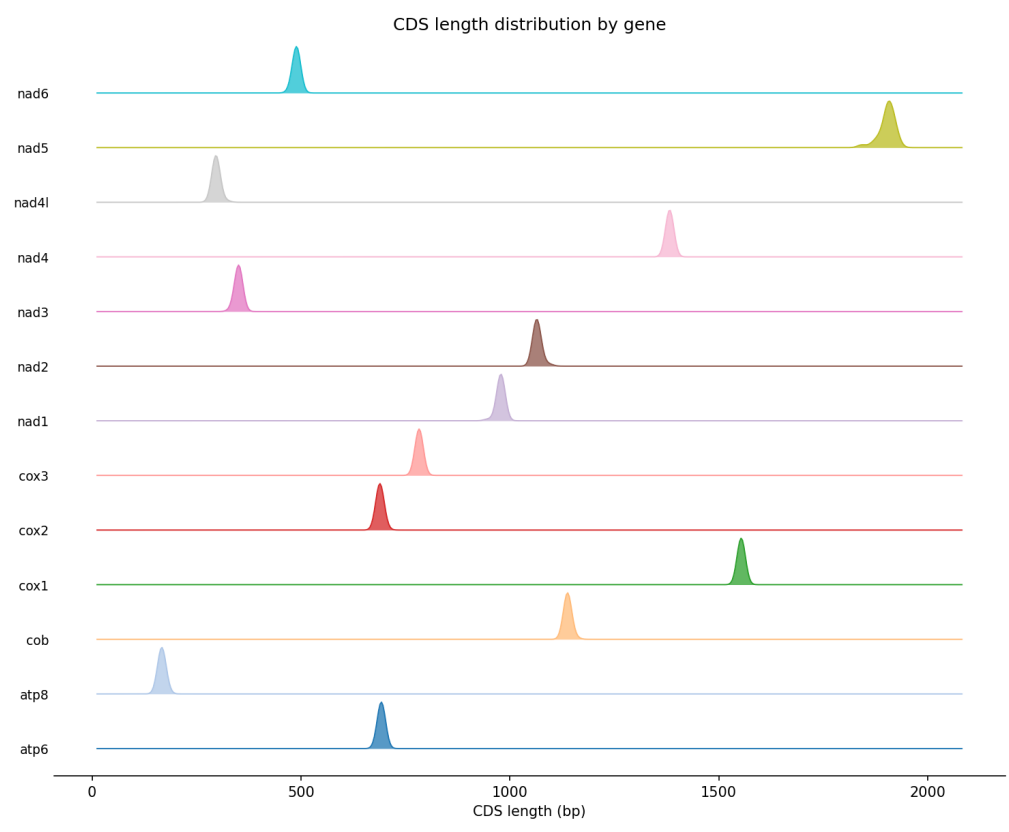
Non-canonical start: 0/30 (0.0%)

Stop codons

Codon	n	%	Status
TAA	15	50.0%	ok
TAG	15	50.0%	ok

Non-canonical stop: 0/30 (0.0%)

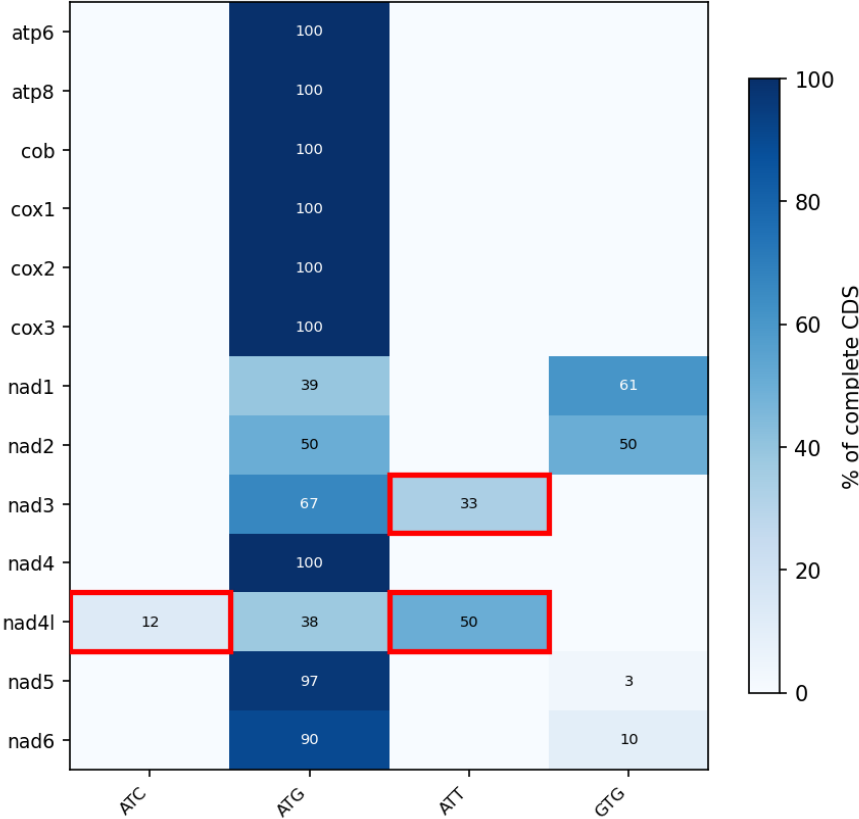
1. CDS Length Distribution



CDS length distribution

2. Start Codon Frequency Heatmap

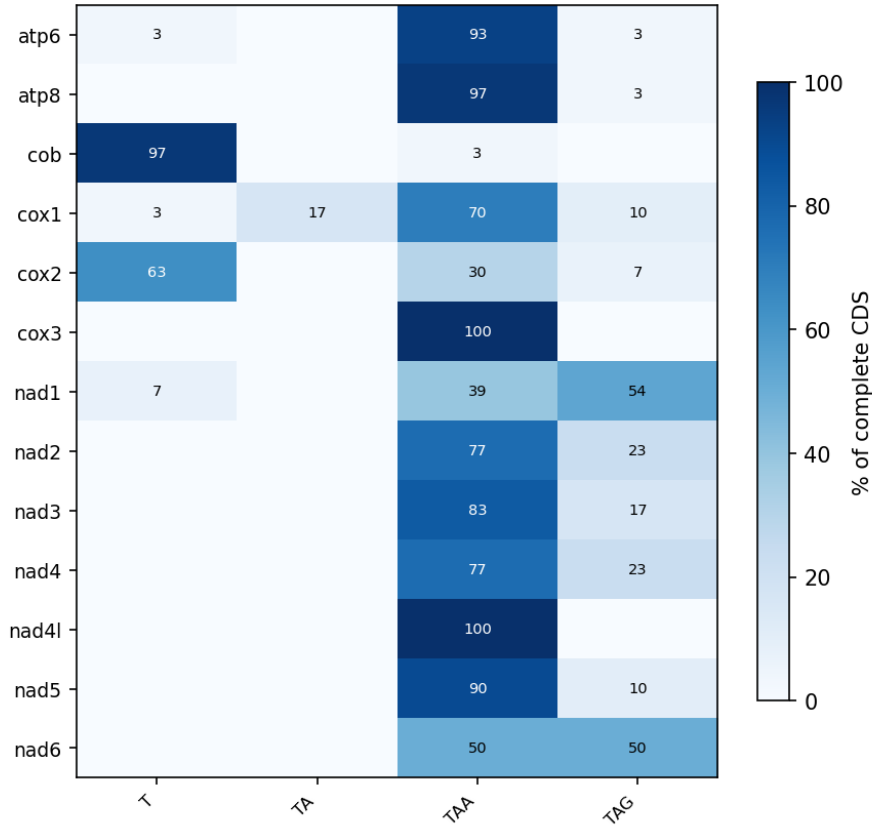
Start codon frequency by gene (complete CDS; red border = non-canonical)



Start codon heatmap

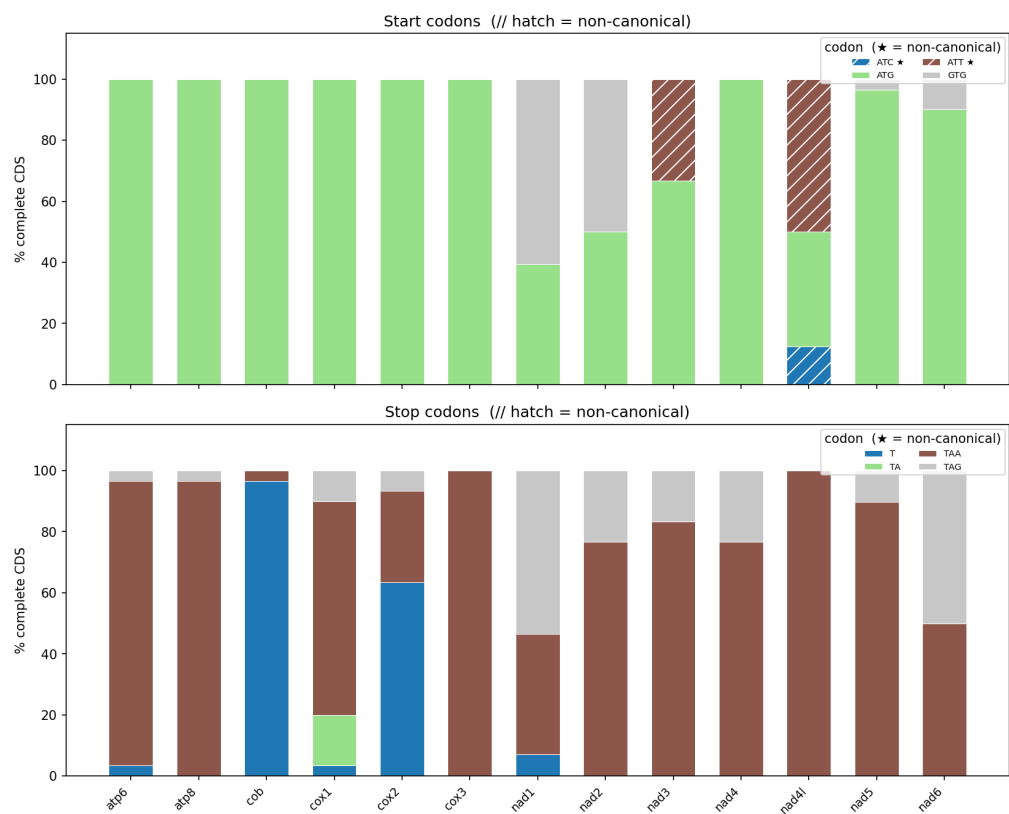
3. Stop Codon Frequency Heatmap

Stop codon frequency by gene (complete CDS; red border = non-canonical)



Stop codon heatmap

4. Start and Stop Codon Distributions



Non-coding Feature Summary

707 non-coding features annotated across 30 record(s).

tRNA

Count per genome (n=30 records): min=19 · max=22 · mean=21.7

Gene	n	min (bp)	max (bp)	mean (bp)	median (bp)
tRNA-Ala	30	68	73	71.1	72.0
tRNA-Arg	30	46	72	68.4	69.0
tRNA-Asn	30	72	73	72.0	72.0
tRNA-Asp	29	69	75	70.4	70.0
tRNA-Cys	30	66	73	69.3	69.0

Gene	n	min (bp)	max (bp)	mean (bp)	median (bp)
tRNA-Gln	30	71	74	71.8	72.0
tRNA-Glu	30	67	73	69.2	69.0
tRNA-Gly	30	67	72	69.3	69.0
tRNA-His	30	67	72	69.2	69.0
tRNA-Ile	30	59	74	71.4	72.0
tRNA-Leu	60	71	74	72.4	72.0
tRNA-Lys	28	57	76	72.3	72.0
tRNA-Met	30	68	75	72.0	72.0
tRNA-Phe	30	50	76	71.1	72.0
tRNA-Pro	30	69	76	70.8	71.0
tRNA-Ser	57	67	73	69.8	70.0
tRNA-Thr	30	69	76	71.7	71.5
tRNA-Trp	27	69	73	69.6	69.0
tRNA-Tyr	30	69	72	70.7	71.0
tRNA-Val	30	62	75	70.6	71.0

tRNA copy number per genome (genes present >1x in at least one record)

Gene	1 copy	2 copies	3+ copies
tRNA-Leu	0	30	0
tRNA-Ser	3	27	0

rRNA

Count per genome (n=27 records): min=2 · max=2 · mean=2.0

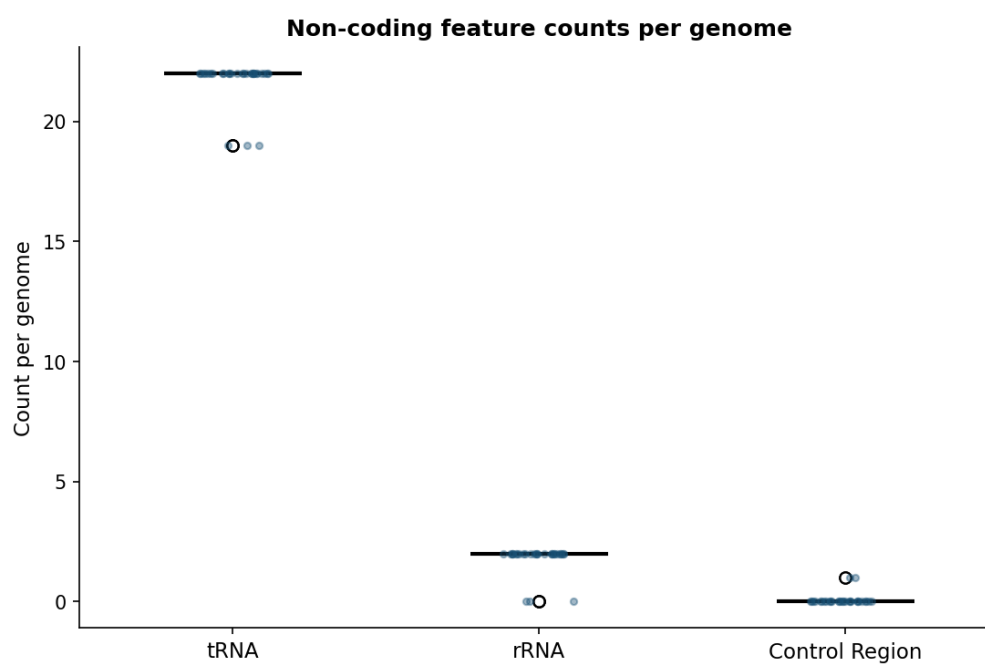
Gene	n	min (bp)	max (bp)	mean (bp)	median (bp)
12S rRNA	27	885	929	902.1	898.0
16S rRNA	27	1531	1646	1596.1	1603.0

Control Region

Count per genome (n=2 records): min=1 · max=1 · mean=1.0

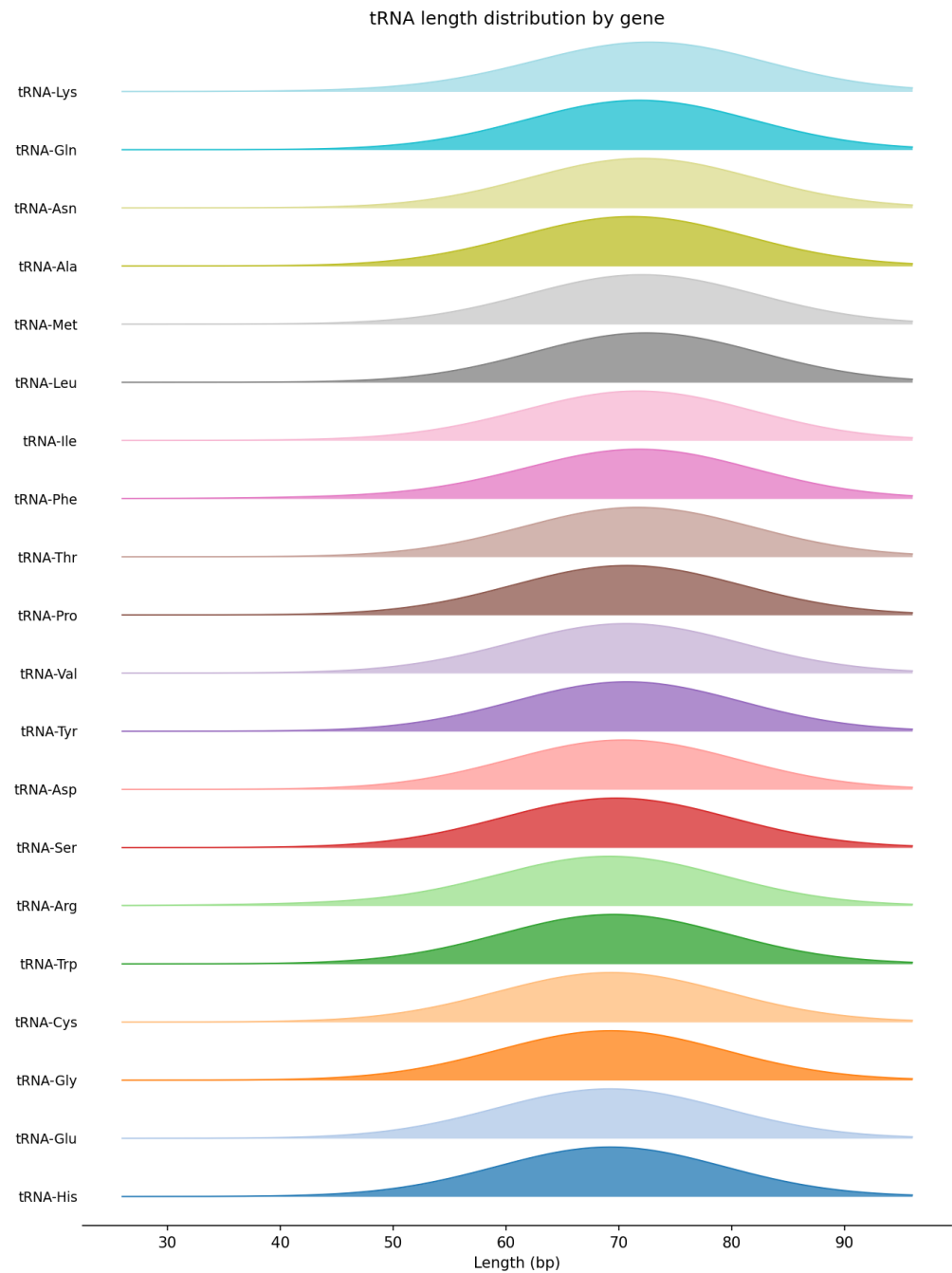
Gene	n	min (bp)	max (bp)	mean (bp)	median (bp)
control region	2	531	552	541.5	541.5

5. Non-coding Feature Counts



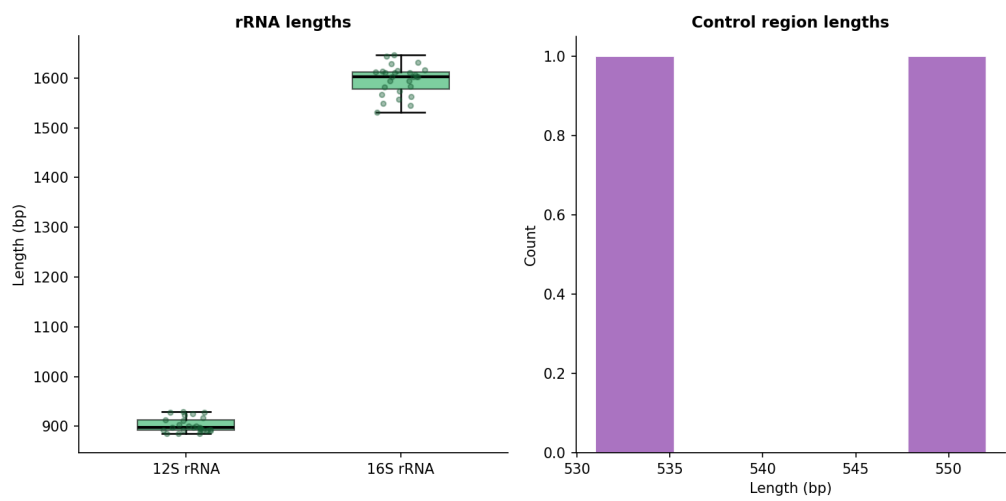
Non-coding feature counts

6. tRNA Length Distribution



tRNA lengths by gene

7. rRNA and Control Region Lengths



rRNA and control region lengths